Alignments practical

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Decoding Genomes B54
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Outline

1. What are alignments?
2. using NCBI
    a. online
    b. on Deigo
    c. for nucleic acid sequences
    d. for amino acid sequences
3. using diamond and mmseqs2 for large datasets
4. using mafft, MUSCLE, Prank and HMMER
5. using trimAL, BMGE, Gblocks
6. Check the impact of alignment and trimming on the inferred phylogeny
7. a small phylogenomics project

Clone the following github repository:

```
git clone https://github.com/oist/b54-alignment-practical.git
```

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1. What are alignments?

- multiple sequence alignments (MSAs)
- input for almost all phylogenomic analyses (tree inference)
- padded sequences
- hypotheses of homology
    - on the level of genes (rows)
    - on the level of positions (columns)

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2. Using NCBI web

Let's search for the Pax6 of Drosophila:

1. Open: https://www.ncbi.nlm.nih.gov/
2. Choose "Gene"
3. Enter "eyeless"

Many hits, do you see "ey" and "toy" in Drosophila melanogaster (fruit fly)?

1. Enter: "43812" into the search field.
2. Scroll down to "Reference assembly".
3. Open GenBank entry: https://www.ncbi.nlm.nih.gov/nuccore/NC_004353.4?from=697689&to=721173&report=genbank
4. Click "Run BLAST"

Spoiler: it will fail.

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2. Using NCBI on Deigo

1. Open GenBank entry: https://www.ncbi.nlm.nih.gov/nuccore/NC_004353.4?from=697689&to=721173&report=genbank
2. Click “FASTA”
3. Select text and copy and paste into a file on Deigo named “drosophila_pax6.fna”
4. Go back x2, scroll down to “eyeless, isoform A” NP_524628.2, click on the accession ID
5. Click “FASTA”
6. Select text and copy and paste into a file on Deigo named “drosophila_pax6_isoformA.faa”

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2. Using NCBI on Deigo contd./2

Open sbatch file: `01_blastn_on_drosohpila_pax6_na.sbatch`

```
#!/bin/bash
[...]
module load ncbi-blast
module load bioinfo-ugrp-modules
module load DB/blastDB/ncbi/2025-12-07-nt

time blastn -db nt -query drosophila_pax6.fna -num_threads 8
-out result_drosophila_pax6.out
```

Run sbatch file:

```
sbatch 01_blastn_on_drosohpila_pax6_na.sbatch
```

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2. Using NCBI on Deigo contd./3

Inspect output: `result_drosophila_pax6.out`

```
                                                                    Score     E
Sequences producing significant alignments:                         (Bits)  Value

AE014135.4 Drosophila melanogaster chromosome 4                      43369   0.0
AC150557.1 Drosophila melanogaster clone BACR06K04, complete sequ...  43369   0.0
AC099309.1 Drosophila melanogaster, chromosome 4, region 101C-101...  43369   0.0
AC010576.16 Drosophila melanogaster, chromosome 4, region 101F-10...  43369   0.0
CP023340.1 Drosophila melanogaster strain rover (forR) chromosome 4   43295   0.0
CP023334.1 Drosophila melanogaster strain sitter (fors) chromosome 4  43258   0.0
CP151524.1 Drosophila melanogaster strain ANU-1 chromosome 4          43081   0.0
[...]
BT011390.1 Drosophila melanogaster GH01157 full insert cDNA           1570    0.0
X79493.1 D.melanogaster ey mRNA (exons 2-9)                           1543    0.0
AF491801.1 Drosophila sechellia strain Clark eyeless gene, intron...  1513    0.0
AF491803.1 Drosophila simulans strain C167 eyeless gene, intron 2...  1507    0.0
AF491797.1 Drosophila sechellia strain 4 eyeless gene, intron 2, ...  1502    0.0
```

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2. Using NCBI on Deigo contd./4

Open sbatch file: `02_blastp_on_drosohpila_pax6_isoformA_aa.sbatch`

```
#!/bin/bash
[...]
module load ncbi-blast
module load bioinfo-ugrp-modules
module load DB/blastDB/uniref90/2020_06

time blastp -db uniref90 -query drosophila_pax6_isoformA.faa
-num_threads 8 -out result_drosophila_pax6_isoformA.out
```

Run sbatch file:

```
sbatch 02_blastp_on_drosohpila_pax6_isoformA_aa.sbatch
```

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2. Using NCBI on Deigo contd./5

Results:

```
                                                                    Score     E
Sequences producing significant alignments:                         (Bits)  Value

UniRef90_O18381 Paired box protein Pax-6 n=22 Tax=melanogaster su...  1620    0.0
UniRef90_A0A6J0G1F7 paired box protein Pax-6 n=5 Tax=melanogaster...  1143    0.0
UniRef90_A0A6P4EXT7 paired box protein Pax-6 isoform X1 n=3 Tax=D...  1113    0.0
UniRef90_UPI0007E6A3E9 paired box protein Pax-6 isoform X1 n=2 Ta...  1091    0.0
UniRef90_A0A1W4V5G2 paired box protein Pax-6 isoform X1 n=2 Tax=D...  1043    0.0
UniRef90_UPI0007E8043C paired box protein Pax-6 isoform X1 n=4 Ta...  1021    0.0
UniRef90_B4H894 Ey n=8 Tax=obscura group TaxID=32355 RepID=B4H894...  974     0.0
UniRef90_A0A3B0K4A9 Blast:Paired box protein Pax-6 n=2 Tax=obscur...  927     0.0
[...]
UniRef90_W8BZH6 Paired box protein Pax-6 n=3 Tax=Ceratitis capita...  580     0.0
UniRef90_A0A0R3NVT5 Ey, isoform C n=2 Tax=Drosophila pseudoobscur...  577     0.0
UniRef90_Q1EBY5 IP14880p (Fragment) n=1 Tax=Drosophila melanogast...  560     0.0
UniRef90_UPI0008112E4E paired box protein Pax-6 isoform X1 n=4 Ta...  543     7e-175
UniRef90_UPI0008DE3598 paired box protein Pax-6 isoform X1 n=4 Ta...  530     9e-170
```

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2. Using NCBI on Deigo contd./6

In summary:

- know which blast program you need (blastn, blastx, blastp)
- know what the query should be (whole gene or just CDS, in na or aa)
- know which database to use (nt, NCBIprot, SwissProt, UnirefXX)

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3. Using Diamond and MMSeqs2

Open sbatch file:

```
03_diamond_on_drosohpila_pax6_isoformA_aa_uniref90.sbatch
#!/bin/bash
[...]
module load bioinfo-ugrp-modules
module load Other/DIAMOND
module load DB/diamondDB/uniref90/2020_06
time diamond blastp --db ${DIAMONDDB}/uniref90.dmnd -q
drosophila_pax6_isoformA.faa -p 16 -o
result_drosophila_pax6_isoformA_diamond_uniref90.out --header
-----------------------------------------------------------
```

Run sbatch file:

```
sbatch 03_diamond_on_drosohpila_pax6_isoformA_aa_uniref90.sbatch
```

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3. Using Diamond and MMSeqs2 contd./2

Results:

Fields: Query ID, Subject ID, Percentage of identical matches, Alignment length, Number of mismatches, Number of gap openings, Start of alignment in query, End of alignment in query, Start of alignment in subject, End of alignment in subject, Expected value, Bit score

# query	subject	ident	length	mism	gapo	q_st	q_end	s_st	s_end	E-value	bitscore
NP_524628.2	UniRef90_O18381	100	801	0	0	38	838	57	857	0.0	1488
NP_524628.2	UniRef90_A0A6P4EXT7	72.9	910	169	12	1	838	1	904	0.0	1147
NP_524628.2	UniRef90_UPI0007E6A3E9	70.8	935	169	13	1	838	1	928	0.0	1126
NP_524628.2	UniRef90_A0A6J0G1F7	71.9	908	180	10	1	838	1	903	0.0	1108
NP_524628.2	UniRef90_UPI0007E8043C	70.7	904	177	13	1	838	1	882	0.0	1106
NP_524628.2	UniRef90_A0A1W4V5G2	68.7	914	190	17	1	838	1	894	0.0	1069
NP_524628.2	UniRef90_B4H894	63.5	921	228	19	1	838	1	896	0.0	970
NP_524628.2	UniRef90_A0A3B0K4A9	61.5	933	241	20	1	838	1	910	0.0	927
NP_524628.2	UniRef90_B4NHW0	61.8	911	252	21	1	838	1	888	2.60e-302	862
NP_524628.2	UniRef90_UPI000A1D07CF	58.6	947	238	28	1	838	1	902	1.15e-278	803
NP_524628.2	UniRef90_B4MF41	54.8	971	241	28	1	838	1	906	4.13e-272	786
NP_524628.2	UniRef90_B3MY40	56.2	921	299	23	1	838	1	900	1.92e-271	784
NP_524628.2	UniRef90_A0A6P4IK30	57.7	953	233	31	1	838	1	898	1.65e-269	779
NP_524628.2	UniRef90_A0A6J2TUG5	55.1	958	253	29	1	838	1	901	4.75e-267	773
NP_524628.2	UniRef90_A0A0M4EQC1	55.8	947	262	30	1	838	1	899	8.17e-265	767
NP_524628.2	UniRef90_A0A6J1LCR4	54.2	946	294	25	1	838	1	915	1.12e-263	765
NP_524628.2	UniRef90_B4L798	54.5	935	284	22	1	838	1	891	2.29e-254	740
NP_524628.2	UniRef90_A0A6P8X7D0	51.2	948	296	31	1	838	1	891	1.11e-235	692
NP_524628.2	UniRef90_A0A6P8KFY1	84.7	430	6	1	1	370	1	430	1.57e-234	672
NP_524628.2	UniRef90_B4R2G4	85.3	423	44	4	1	411	1	417	9.15e-227	652
NP_524628.2	UniRef90_UPI0007E8880A	74.3	495	57	7	1	428	1	492	1.75e-223	646
NP_524628.2	UniRef90_A0A0R3NVT5	67.1	493	90	8	1	428	1	486	5.77e-189	558
NP_524628.2	UniRef90_W8BZH6	44.9	984	334	31	38	838	134	1092	4.12e-183	564
NP_524628.2	UniRef90_Q1EBY5	100	277	0	0	38	314	81	357	2.20e-181	533
NP_524628.2	UniRef90_UPI0008DE3598	43.8	1001	338	32	38	838	141	1116	9.29e-177	548

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3. Using Diamond and MMSeqs2 contd./3

Open sbatch file:

```
04_diamond_on_drosohpila_pax6_isoformA_aa_ncbinr.sbatch
#!/bin/bash
[...]
module load bioinfo-ugrp-modules
module load Other/DIAMOND
module load DB/diamondDB/ncbi/2022-07
time diamond blastp --db ${DIAMONDDB}/nr.dmnd -q
drosophila_pax6_isoformA.faa -p 16 -o
result_drosophila_pax6_isoformA_diamond_ncbinr.out --header
-----------------------------------------------------------
```

Run sbatch file:

```
sbatch 04_diamond_on_drosohpila_pax6_isoformA_aa_ncbinr.sbatch
```

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3. Using Diamond and MMSeqs2 contd./4

Results:

```
# Fields: Query ID, Subject ID, Percentage of identical matches, Alignment length, Number of mismatches, Number of gap openings, Start of alignment in query, End of alignment in query, Start of alignment in subject, End of alignment in subject, Expected value, Bit score
```

# query	subject	ident	length	mism	gapo	q_st	q_end	s_st	s_end	E-value	bitscore
NP_524628.2	NP_524628.2	100	838	0	0	1	838	1	838	0.0	1561
NP_524628.2	CAA56038.1	99.8	838	2	0	1	838	1	838	0.0	1559
NP_524628.2	NP_001014693.1	93.2	898	1	1	1	838	1	898	0.0	1531
NP_524628.2	XP_016037331.1	93.6	839	53	1	1	838	1	839	0.0	1489
NP_524628.2	O18381.3	100	801	0	0	38	838	57	857	0.0	1488
NP_524628.2	NP_001014694.1	100	801	0	0	38	838	57	857	0.0	1488
NP_524628.2	XP_033168158.1	92.6	839	61	1	1	838	1	839	0.0	1470
NP_524628.2	XP_032579804.1	92.5	839	62	1	1	838	1	839	0.0	1469
NP_524628.2	XP_016037329.1	87.2	899	54	2	1	838	1	899	0.0	1459
NP_524628.2	XP_033168155.1	86.3	899	62	2	1	838	1	899	0.0	1440
NP_524628.2	XP_002043695.1	86.2	899	63	2	1	838	1	899	0.0	1439
NP_524628.2	XP_033168156.1	86.1	887	62	2	13	838	6	892	0.0	1417
NP_524628.2	XP_016037332.1	93.3	802	53	1	38	838	57	858	0.0	1416
NP_524628.2	XP_033168157.1	92.3	802	61	1	38	838	57	858	0.0	1397
NP_524628.2	XP_039496473.1	83.4	841	134	4	1	838	1	838	0.0	1291
NP_524628.2	XP_043657339.1	84.7	841	125	3	1	838	1	840	0.0	1276
NP_524628.2	XP_015008645.1	84.1	841	130	3	1	838	1	840	0.0	1263
NP_524628.2	XP_039496471.1	77.7	901	135	5	1	838	1	898	0.0	1261

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3. Using Diamond and MMSeqs2 contd./5

In summary:

- we didn’t use mmseqs, because there is no database on Deigo for it
- also, usually we would use these tools to do all against all searches between genomes, to identify homologous genes, i.e. the database would be created from the dataset
- let’s use the prepared fasta file for the downstream analysis “pax6_family.fasta”

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4. Using MAFFT, MUSCLE, PRANK and Clustal

Open sbatch file: `06_mafft_linsi.sbatch`

```
#!/bin/bash
[...]
module load bioinfo-ugrp-modules
# do not load DebianMed, because its mafft version is broken
module load mafft

time linsi --thread 16 pax6_family.fasta >
pax6_family.linsi.fasta
```

Run sbatch file:

```
sbatch 06_mafft_linsi.sbatch
```

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4. Using MAFFT, MUSCLE, PRANK and Clustal contd./2

Open sbatch file: `07_clustalw.sbatch`

```
#!/bin/bash
[...]
module load bioinfo-ugrp-modules
module load DebianMed
module load clustalw

time clustalw -infile=pax6_family.fasta
-outfile=pax6_family.clustalw.fasta
```

Run sbatch file:

```
sbatch 07_clustalw.sbatch
```

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4. Using MAFFT, MUSCLE, PRANK and Clustal contd./3

Open sbatch file: `08_muscle.sbatch`

```
#!/bin/bash
[...]
module load bioinfo-ugrp-modules
module load DebianMed
module load muscle

time muscle -align pax6_family.fasta -output
pax6_family.muscle.fasta
```

Run sbatch file:

```
sbatch 08_muscle.sbatch
```

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4. Using MAFFT, MUSCLE, PRANK and Clustal contd./4

Open sbatch file: `09_prank.sbatch`

```
#!/bin/bash
[...]
module load bioinfo-ugrp-modules
module load DebianMed
module load prank

time prank -d=pax6_family.fasta -o=pax6_family.prank.fasta
```

Run sbatch file:

```
sbatch 09_prank.sbatch
```

OIST

4. Using MAFFT, MUSCLE, PRANK and Clustal contd./5

Open file: `10_alignment_statistics.py`

Run the python script:

```
./10_alignment_statistics.py
```

Results:

```
{'label': 'pax6_family.clustalw.fasta',
'ncol': 859, 'gaps': 2284, 'gap_pct': 44.32, 'conserved': 92}
{'label': 'pax6_family.linsi.fasta',
'ncol': 892, 'gaps': 2482, 'gap_pct': 46.38, 'conserved': 91}
{'label': 'pax6_family.muscle.fasta',
'ncol': 922, 'gaps': 2662, 'gap_pct': 48.12, 'conserved': 92}
{'label': 'pax6_family.prank.fasta.best.fas',
'ncol': 991, 'gaps': 3076, 'gap_pct': 51.73, 'conserved': 92}
```

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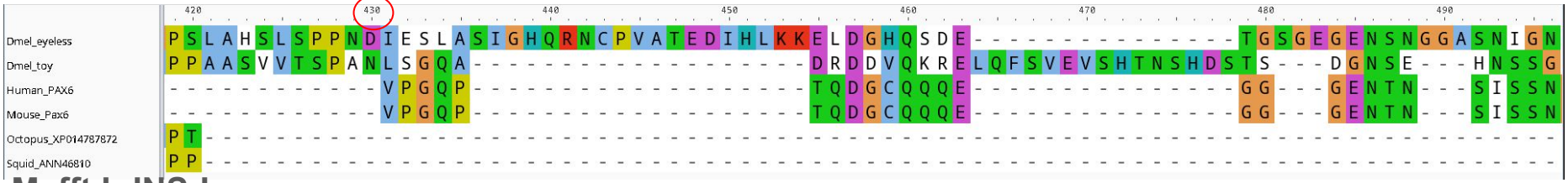
4. Using MAFFT, MUSCLE, PRANK and Clustal contd./6

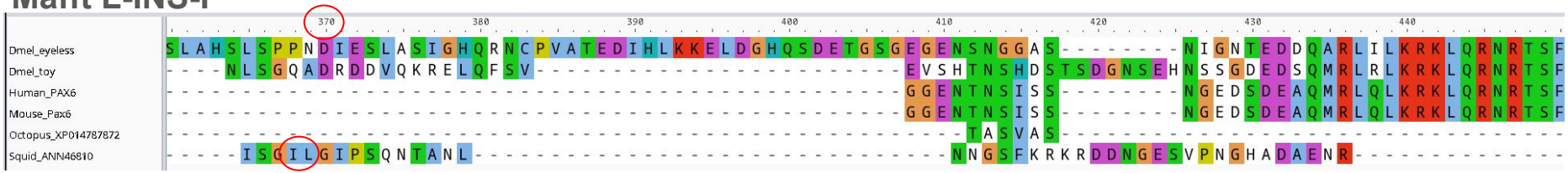
Let's inspect alignments in AliView: https://ormbunkar.se/aliview/

Set color scheme to ClustalX: https://www.jalview.org/help/html/colourSchemes/clustal.html

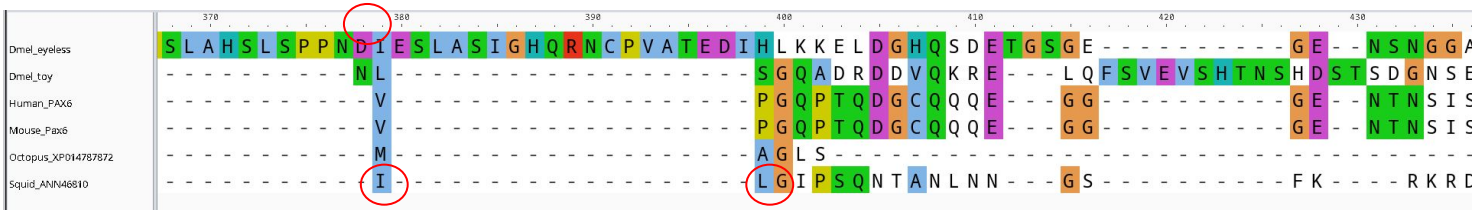
Prank



Mafft L-INS-I



Muscle



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5. Using trimAl and BMGE

Installing trimAl
https://vicfero.github.io/trimal/

```
git clone https://github.com/inab/trimal.git
salloc -p short -c 4 --mem 10G --time 0-1:00:00
cd trimal/source
make & cd ../..
```

Installing BMGE
https://research.pasteur.fr/en/software/bmge-block-mapping-and-gathering-with-entropy/

```
git clone https://gitlab.pasteur.fr/GIPhy/BMGE.git
module load java-jdk/11
cd BMGE/src
javac BMGE.java
echo Main-Class: BMGE > MANIFEST.MF
jar -cmvf MANIFEST.MF BMGE.jar BMGE.class bmge/*.class
```

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5. Using trimAl and BMGE contd./2

Open sbatch file: `11_trimal.sbatch`

```
#!/bin/bash
[...]

time trimal/source/trimal -in alignment.fasta -out
alignment.fasta.trimal -gappyout
```

Run sbatch file:

```
sbatch 11_trimal.sbatch
```

Note: ClustalW alignment dies with Segmentation fault

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5. Using trimAl and BMGE contd./3

Open sbatch file: `12_BMGE.sbatch`

```
#!/bin/bash
[...]

module load java-jdk/11

time java -jar BMGE/src/BMGE.jar -i alignment.fasta -t AA -o
alignment.fasta.BMGE -oh alignment.fasta.BMGE.html
```

Run sbatch file:

```
sbatch 12_BMGE.sbatch
```

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5. Using trimAl and BMGE contd./3

Run python script: `./13_trimmed_alignment_statistics.py`

```
TrimAL
TrimAL
{'label': 'pax6_family.linsi.fasta.trimal',
'ncol': 419, 'gaps': 342, 'gap_pct': 13.60, 'conserved': 91}
{'label': 'pax6_family.muscle.fasta.trimal',
'ncol': 353, 'gaps': 167, 'gap_pct':  7.89, 'conserved': 92}
{'label': 'pax6_family.prank.fasta.best.fas.trimal',
'ncol': 431, 'gaps': 469, 'gap_pct': 18.14, 'conserved': 92}
BMGE
{'label': 'pax6_family.linsi.fasta.BMGE',
'ncol': 496, 'gaps': 573, 'gap_pct': 19.25, 'conserved': 91}
{'label': 'pax6_family.muscle.fasta.BMGE',
'ncol': 443, 'gaps': 393, 'gap_pct': 14.79, 'conserved': 92}
{'label': 'pax6_family.prank.fasta.best.fas.BMGE',
'ncol': 427, 'gaps': 472, 'gap_pct': 18.42, 'conserved': 92}
```

Before trimming:

```
linsi:  ncol: 892, gaps: 2482, gap_pct: 46.38, conserved: 91}
muscle: ncol: 922, gaps: 2662, gap_pct: 48.12, conserved: 92}
prank:  ncol: 991, gaps: 3076, gap_pct: 51.73, conserved: 92}
```

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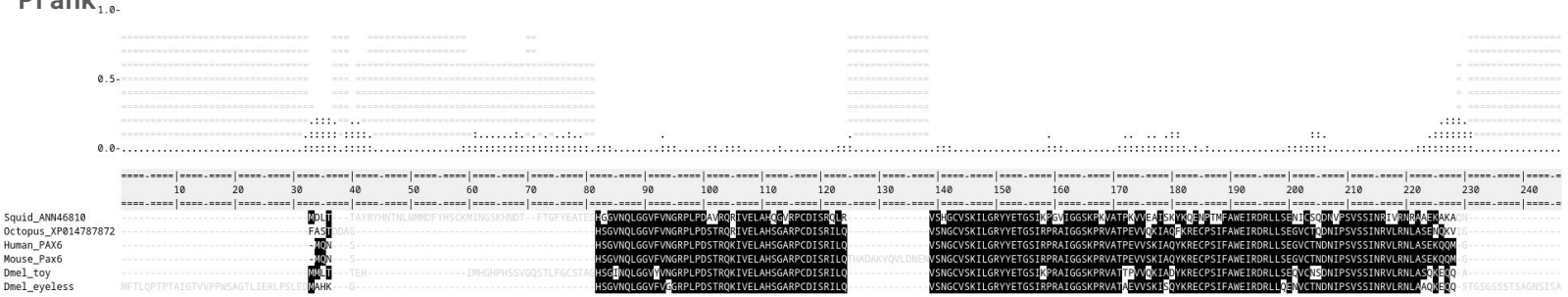
5. Using trimAl and BMGE contd./3

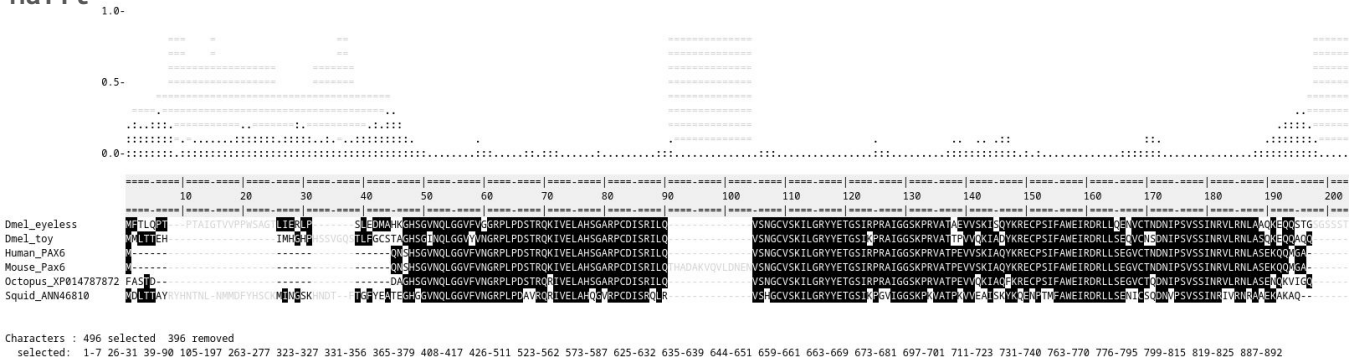
BMGE entropy profiles

Prank



Characters : 427 selected 564 removed
selected: 33-36 82-124 139-228 308-322 383-388 392-397 411-420 431-435 455-463 484-488 492-500 504-581 594-598 627-637 645-648 654-661 668-678 692-694 701-705 710-714 739-747 763-767 770-773 779-784 793-796 845-861 864-888 899-901 917-925 934-940 970-972 989-991

Mafft



Characters : 496 selected 396 removed
selected: 1-7 26-31 39-90 105-197 263-277 323-327 331-356 365-379 408-417 426-511 523-562 573-587 625-632 635-639 644-651 659-661 663-669 673-681 697-701 711-723 731-740 763-770 776-795 799-815 819-825 887-892

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6. Checking impact of align. and trimming on phylogeny

Open sbatch file: `14_infer_trees.sbatch`

```
#!/bin/bash
[...]

module load bioinfo-ugrp-modules
module load Other/iqtree

time iqtree -T 4 -s alignment.fasta -m LG+G
```

Run sbatch file:

```
sbatch 14_infer_trees.sbatch
```

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6. Checking impact of align. and trim. on phylo. contd/2

ClustalW

```
(Human_PAX6:0.0000010000,Mouse_Pax6:0.0000010000,((Dmel_toy:0.43,Dmel_eyeless:0.93):0.13,
(Octopus_XP014787872:0.19,Squid_ANN46810:3.02):0.67):0.3);
```

Linsi

```
(Dmel_eyeless:0.86,Dmel_toy:0.37,((Human_PAX6:0.0000010000,Mouse_Pax6:0.0000010000):0.32,
(Octopus_XP014787872:0.17,Squid_ANN46810:2.16):0.31):0.11);
```

Linsi BMGE

```
(Dmel_eyeless:0.84,Dmel_toy:0.36,((Human_PAX6:0.0000010000,Mouse_Pax6:0.0000010000):0.32,
(Octopus_XP014787872:0.18,Squid_ANN46810:2.15):0.30):0.11);
```

Linsi trimAl

```
(Dmel_eyeless:0.66,Dmel_toy:0.31,((Human_PAX6:0.0000010000,Mouse_Pax6:0.0000010000):0.30,
(Octopus_XP014787872:0.17,Squid_ANN46810:1.98):0.28):0.11);
```

Muscle

```
(Octopus_XP014787872:0.30,Squid_ANN46810:2.13,((Human_PAX6:0.0000010000,Mouse_Pax6:0.0000010000):0.16,
(Dmel_eyeless:0.72,Dmel_toy:0.55):0.16):0.35);
```

Muscle BMGE

```
(Octopus_XP014787872:0.23,Squid_ANN46810:2.07,((Human_PAX6:0.0000010000,Mouse_Pax6:0.0000010000):0.15,
(Dmel_eyeless:0.66,Dmel_toy:0.48):0.16):0.37);
```

Muscle trimAl

```
(Octopus_XP014787872:0.25,Squid_ANN46810:2.01,((Human_PAX6:0.0000010000,Mouse_Pax6:0.0000010000):0.15,
(Dmel_eyeless:0.57,Dmel_toy:0.52):0.15):0.34);
```

Prank

```
(Squid_ANN46810:1.74,(Octopus_XP014787872:0.35,Dmel_eyeless:0.47):0.07,
((Human_PAX6:0.0000010000,Mouse_Pax6:0.0000010000):0.26,Dmel_toy:0.40):0.03);
```

Prank BMGE

```
(Squid_ANN46810:1.25,((Octopus_XP014787872:0.34,(Human_PAX6:0.0000010000,Mouse_Pax6:0.0000010000):0.17):0.09,
Dmel_toy:0.33):0.10,Dmel_eyeless:0.37);
```

Prank trimAl

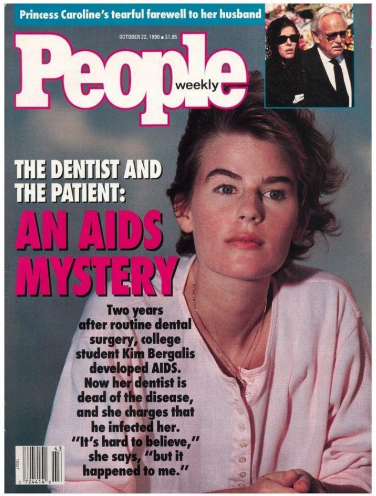
```
(Squid_ANN46810:1.26,((Octopus_XP014787872:0.34,(Human_PAX6:0.0000010000,Mouse_Pax6:0.0000010000):0.19):0.10,
Dmel_toy:0.33):0.09,Dmel_eyeless:0.37);
```

6. Phylogenomics project: Tracing infection by HIV

Molecular Epidemiology of HIV Transmission in a Dental Practice

CHIN-YIH OU, CAROL A. CIESIELSKI, GERALD MYERS, CLAUDIU I. BANDEA, CHI-CHENG LUO, BETTE T. M. KORBER, JAMES I. MULLINS, GERALD SCHOCHETMAN, RUTH L. BERKELMAN, A. NIKKI ECONOMOU, JOHN J. WITTE, LAWRENCE J. FURMAN, GLEN A. SATTEN, KERSTI A. MACLNNES, JAMES W. CURRAN, HAROLD W. JAFFE, LABORATORY INVESTIGATION GROUP, AND EPIDEMIOLOGIC INVESTIGATION GROUP fewer Authors

SCIENCE • 22 May 1992 • Vol 256, Issue 5060 • pp. 1165-1171 • DOI: 10.1126/science.256.5060.1165

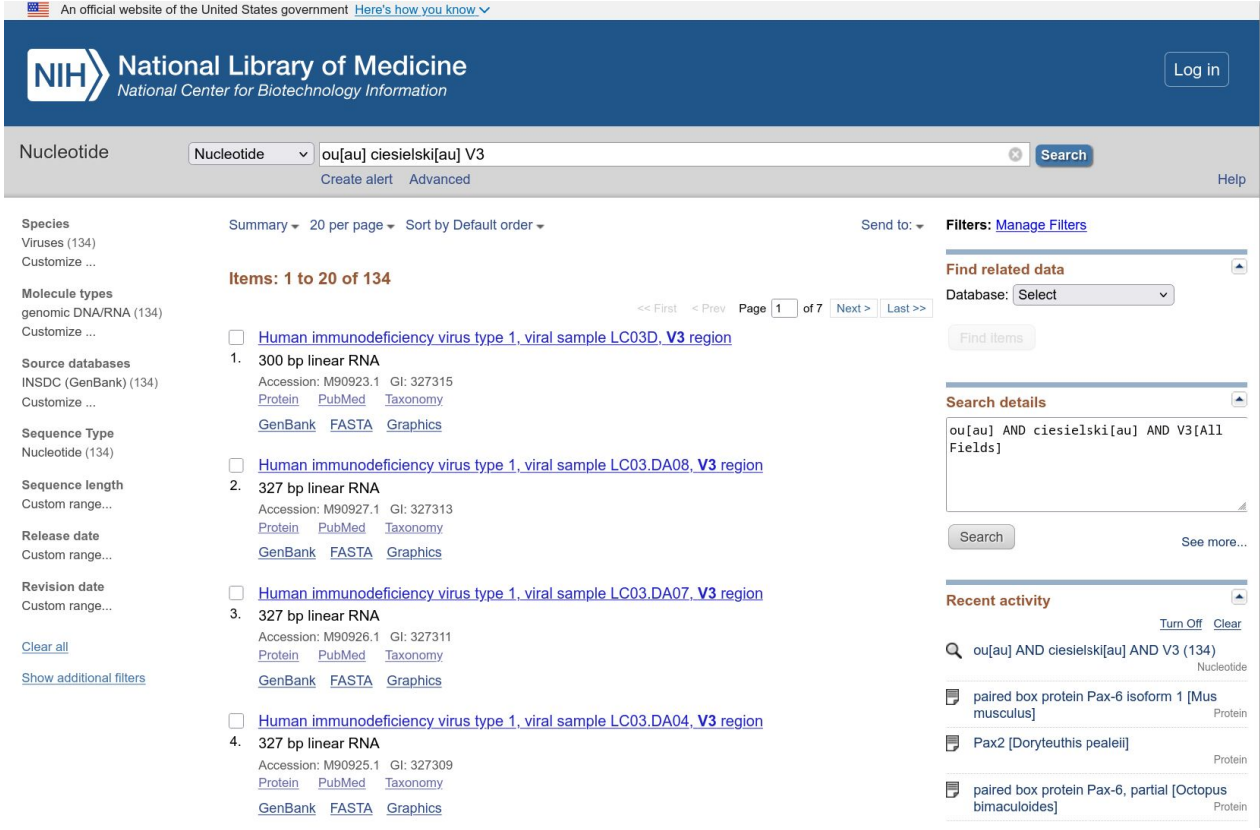


https://doi.org/10.1126/science.256.5060.1165

6. Phylogenomics project: Download the data

https://www.ncbi.nlm.nih.gov/nuccore/?term=ou%5Bau%5D%20ciesielski%5Bau%5D%20V3

- download dentist's sequences: FLD1,2,4,5,7,8
- patient's sequences: FLP[A-H]: take 3 isolates/patient
- Local controls (LCXX, FLQ): take at least 10



OIST

https://doi.org/10.1126/science.256.5060.1165

6. Phylogenomics project: Download the data

Open file: `15_download_hiv_samples.py`

```
# Write your email address below:
Entrez.email = "EMAIL ADDRESS"
```

Save.

Run python script: `./15_download_hiv_samples.py`

https://doi.org/10.1126/science.256.5060.1165

6. Phylogenomics project: Align samples

Open sbatch file: `16_mafft_hiv.sbatch`

```
module load bioinfo-ugrp-modules
# do not load DebianMed, because its mafft version is broken
module load mafft

time mafft --thread 4 --auto hiv_samples.fasta >
hiv_samples.mafft.fasta
```

Run sbatch file: sbatch `16_mafft_hiv.sbatch`

OIST

https://doi.org/10.1126/science.256.5060.1165

6. Phylogenomics project: Infer phylogeny

Open sbatch file: `17_iqtree_hiv.sbatch`

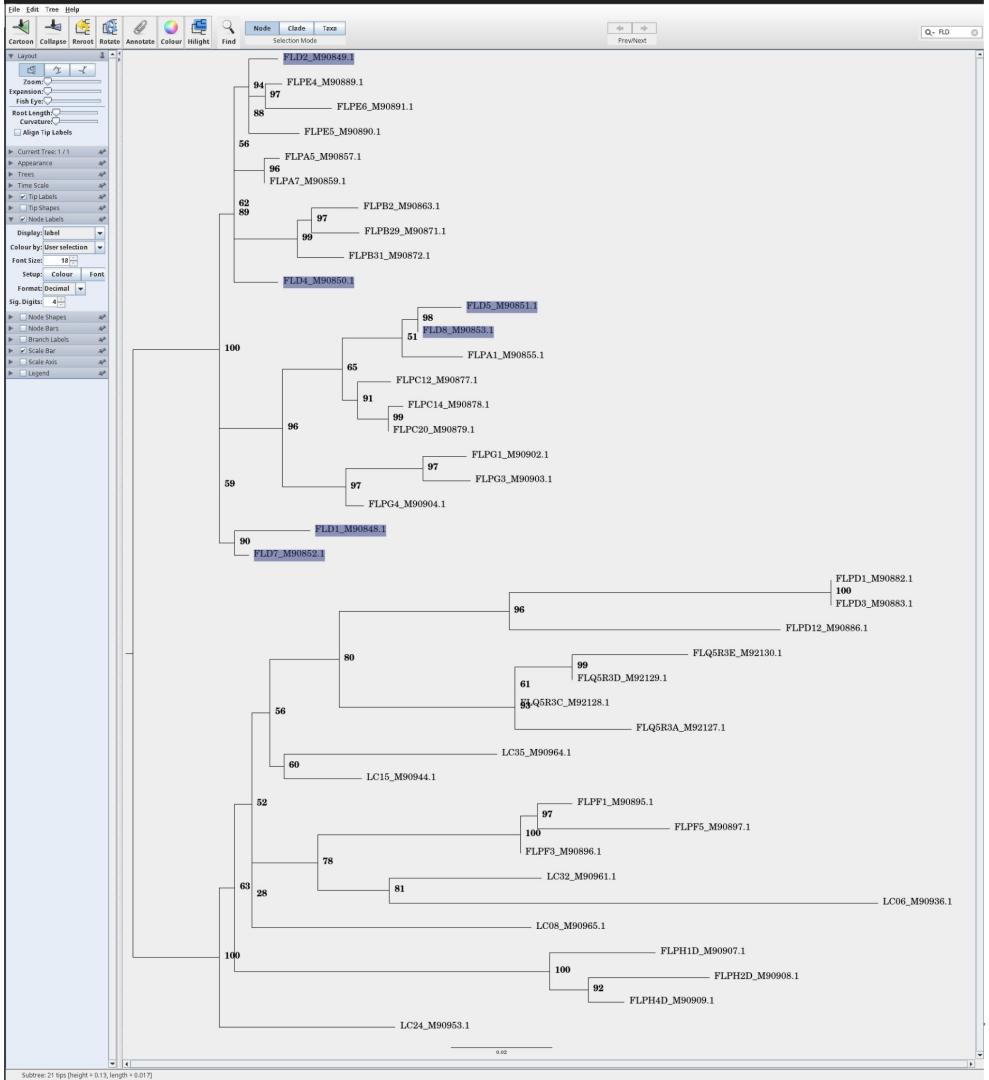
```
module load bioinfo-ugrp-modules
module load Other/iqtree

time iqtree -T 4 -s hiv_samples.mafft.fasta -m MFP -B 1000
```

Run sbatch file: sbatch `17_iqtree_hiv.sbatch`

https://doi.org/10.1126/science.256.5060.1165

6. Phylogenomics project: Inspect phylogeny

Download Figtree:
https://tree.bio.ed.ac.uk/software/figtree/
Download TreeViewer:
https://treeviewer.org/



OIST

https://doi.org/10.1126/science.256.5060.1165